

MATH3085 Data Analysis Project

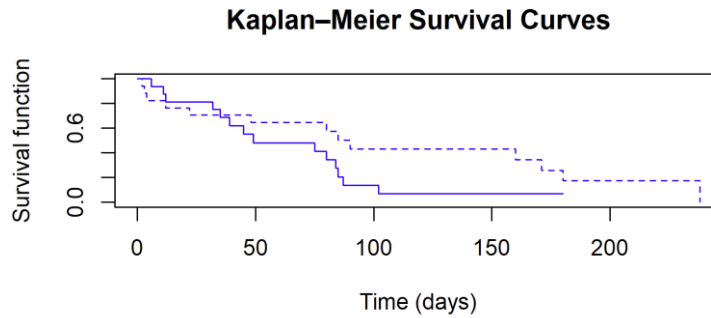
Q1. a) Let the order of the observed relapses within a treatment group be: $t_1 < t_2 < \dots < t_k$.

Let d_j denote the number of relapses at time t_j and n_j denote the number of individuals at risk.

The Kaplan–Meier estimator of the survival function is given by: $\hat{S}(t) = \prod_{\{t_j \leq t\}} 1 - \frac{d_j}{n_j}$

This estimator is then computed separately for each treatment group.

From the R code, we have the following graph:



From the Kaplan–Meier survival curves, the triple-drug treatment group consistently shows higher survival probabilities while two-drug group shows a sharper and earlier drop in its survival curve, indicating that patients on this regimen tend to relapse sooner.

b) The estimated probability that relapses occur later than 120 days after treatment is given by:

$$\hat{S}(120) = \prod_{\{t_j \leq 120\}} 1 - \frac{d_j}{n_j}$$

The variance of this Kaplan–Meier estimator is estimated using Greenwood's formula:

$$\text{Var} [\hat{S}(120)]^2 = [\hat{S}(120)]^2 \times \sum_{\{t_j \leq 120\}} \frac{d_j}{n_j \times (n_j - d_j)}$$

And thus, the estimated 95% confidence interval for $S(120)$ is then given by:

$$\hat{S}(120) \pm 1.96 \times \sqrt{\text{Var} \hat{S}(120)}$$

From the R code, we have the results:

```
call: survfit(formula = Surv(time, status) ~ treatment, data = data)

              treatment=Triple-drug
      time      n.risk    n.event   survival    std.err
120.000         5.000      9.000      0.431      0.128
lower 95% CI upper 95% CI
0.241         0.771

              treatment=Two-drug
      time      n.risk    n.event   survival    std.err
120.0000         1.0000     14.0000    0.0688     0.0662
lower 95% CI upper 95% CI
0.0104         0.4544
```

The estimated probability of remaining relapse-free beyond 120 days was 0.431 for patients receiving the triple-drug treatment, with a corresponding 95% confidence interval of (0.241, 0.771). In contrast, the probability of surviving past 120 days in the two-drug treatment group was substantially lower at 0.0688, with a 95% confidence interval of (0.0104, 0.4544). These results suggest that patients treated with the triple-drug treatment have a higher probability of not experiencing CD4 count relapses after 120 days compared with those treated with the two-drug treatment.

c) Let T_i denote the relapse time for the i -th patient and let x_i denote the treatment indicator such that $x_i = 0$ patients receiving the triple-drug treatment and $x_i = 1$ or patients receiving the two-drug treatment. The Cox proportional hazards model assumes that the hazard function for T_i is given by: $h_{T_i}(t) = h_0(t) \times \exp(x_i^T \times \beta)$

where $h_0(t)$ is the baseline hazard function and β is the regression coefficient corresponding to treatment.

The survival function and the baseline survival function are respectively given by:

$$S_{T_i}(t) = [S_0(t)]^{\exp(x_i^T \times \beta)} \text{ \& \ } S_0(t) = \exp\left(-\int_0^t h_0(s) ds\right)$$

where the baseline survivor function, $S_0(t)$, is corresponding to the triple-drug treatment group.

The hazard ratio is given by: $h_0(t) = \frac{h(t|x_i=1)}{h(t|x_i=0)} = \exp(\beta)$ where $\exp(\beta) > 1$ indicates that the hazard of relapses is higher for the two-drug treatment than for the triple-drug treatment and vice versa. If $\exp \beta = 1$, and the hazards for both treatments are equal.

The R code generated following results and graphs:

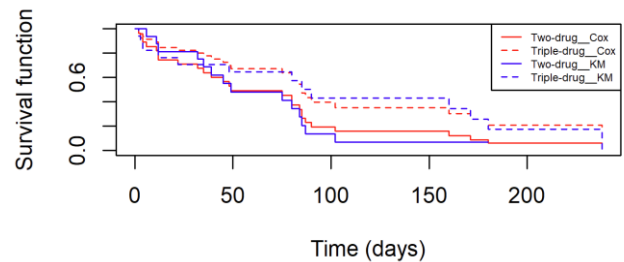
```
> summary(cox_model)
Call:
coxph(formula = Surv(time, status) ~ treatment, data = data)

n= 34, number of events= 27

              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentTwo-drug  0.5728   1.7731  0.4052  1.414  0.157

              exp(coef) exp(-coef) lower .95 upper .95
treatmentTwo-drug      1.773      0.564   0.8014   3.923

Concordance= 0.55 (se = 0.06 )
Likelihood ratio test= 2.01 on 1 df,  p=0.2
Wald test               = 2 on 1 df,  p=0.2
Score (logrank) test = 2.05 on 1 df,  p=0.2
```



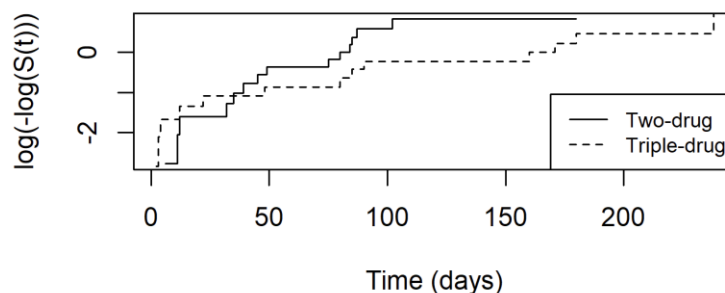
Under this model, the hazard ratio is 1.773, indicating that patients receiving the two-drug treatment have an increased instantaneous hazard compared with those receiving the triple-drug treatment. The Cox model suggests that patients on the triple-drug treatment experience a higher instantaneous risk of death than those on the two-drug regimen. Therefore, the two-drug treatment is associated with better survival, while the triple-drug treatment increases the hazard. Both the KM curves and the Cox model indicate the same result that the triple-drug treatment is associated with a higher hazard and worse survival compared to the two-drug treatment.

d) To check whether proportional hazards assumption seem reasonable, we first compute $\log \{-\log [S(t)]\}$ survival curves. Under the proportional hazards' assumption, these transformed curves for different groups should be parallel because:

$$\log \{-\log [S(t)]\} = \log \{-\exp(x_i^T \times \beta) \times \log [S_0(t)]\} = x_i^T \times \beta + \log \{-\log [S_0(t)]\}$$

which implies that the curves differ only by a constant vertical shift.

From the R code, we have the following graph:



Based on the plot, the proportional hazards assumption appears reasonable as the two treatment groups' curves are roughly parallel.

Q2. a) The full Cox model included treatment, age, and risk group as explanatory variables:

$$h(t) = h_0(t) \times \exp(\beta_1 \times \text{treatment} + \beta_2 \times \text{age} + \beta_3 \times \text{riskgpp})$$

where $h_0(t)$ is the baseline hazard function

```
> summary(DRS_cox_model)
Call:
coxph(formula = Surv(time, status) ~ treatment + age + riskgp,
      data = DRS_data)

n= 197, number of events= 86

              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentplacebo 0.518693  1.679831 0.218009 2.379  0.0173 *
age              0.002229  1.002232 0.007176 0.311  0.7561
riskgp          0.118926  1.126286 0.073718 1.613  0.1067
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
treatmentplacebo    1.680    0.5953    1.0957    2.575
age                 1.002    0.9978    0.9882    1.016
riskgp              1.126    0.8879    0.9748    1.301

Concordance= 0.585  (se = 0.033 )
Likelihood ratio test= 7.79  on 3 df,  p=0.05
Wald test            = 7.81  on 3 df,  p=0.05
Score (logrank) test = 7.91  on 3 df,  p=0.05
```

```
> summary(DRS_preferred_cox_model)
Call:
coxph(formula = Surv(time, status) ~ treatment + riskgp, data = DRS_data)

n= 197, number of events= 86

              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentplacebo 0.52351  1.68794 0.21740 2.408  0.016 *
riskgp          0.11807  1.12532 0.07361 1.604  0.109
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
treatmentplacebo    1.688    0.5924    1.1023    2.585
riskgp              1.125    0.8886    0.9741    1.300

Concordance= 0.586  (se = 0.033 )
Likelihood ratio test= 7.7  on 2 df,  p=0.02
Wald test            = 7.72  on 2 df,  p=0.02
Score (logrank) test = 7.82  on 2 df,  p=0.02
```

The full Cox proportional hazards model includes treatment, age, and risk group as covariates. The model output from R shows that treatment was statistically significant, with a p-value of 0.017, while age and risk group were less significant, with higher p-values of 0.756 and 0.107, respectively. Given the very large p-value for age, this variable should be removed from the model. Treatment is retained due to its clear statistical significance, and risk group should also be retained as its p-value is close to commonly used significance thresholds (5% or 10%) and excluding it would result in a simple model with only one covariate.

Therefore, the preferred model is given by:

$$h(t) = h_0(t) \times \exp(\beta_1 \times \text{treatment} + \beta_2 \times \text{riskgpp}) \text{ where } h_0(t) \text{ is the baseline hazard function}$$

The preferred model therefore now includes treatment and risk group. Under this model, patients receiving placebo have an estimated 68.8% higher hazard of blindness compared with those receiving laser treatment, and each one-unit increase in risk group is associated with an approximate 12.5% increase in hazard.

b) The R code generated following results:

```
> any(duplicated(event_times))
[1] TRUE
```

```
> summary(cox_efron)
Call:
coxph(formula = Surv(time, status) ~ treatment + riskgp, data = DRS_data,
      ties = "efron")

n= 197, number of events= 86

              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentplacebo 0.52351   1.68794  0.21740  2.408  0.016 *
riskgp           0.11807   1.12532  0.07361  1.604  0.109
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
treatmentplacebo    1.688    0.5924    1.1023    2.585
riskgp              1.125    0.8886    0.9741    1.300

Concordance= 0.586 (se = 0.033 )
Likelihood ratio test= 7.7  on 2 df,  p=0.02
Wald test              = 7.72  on 2 df,  p=0.02
Score (logrank) test = 7.82  on 2 df,  p=0.02
```

```
> summary(cox_exact)
Call:
coxph(formula = Surv(time, status) ~ treatment + riskgp, data = DRS_data,
      ties = "exact")

n= 197, number of events= 86

              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentplacebo 0.53442   1.70646  0.22051  2.424  0.0154 *
riskgp           0.12162   1.12932  0.07482  1.625  0.1041
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
treatmentplacebo    1.706    0.5860    1.1076    2.629
riskgp              1.129    0.8855    0.9753    1.308

Concordance= 0.586 (se = 0.033 )
Likelihood ratio test= 7.82  on 2 df,  p=0.02
Wald test              = 7.83  on 2 df,  p=0.02
Score (logrank) test = 7.94  on 2 df,  p=0.02
```

The output from R shows that duplicate survival times exist in this dataset. The preferred model from part (a), including the treatment group and the risk group, is calibrated using both the exact partial probability method and the Efron approximation method. The results of the two methods are very similar. Using the exact method, the estimated risk ratio for placebo versus laser treatment is 1.71, whereas the Efron approximation yields 1.69. In both cases, the treatment variable is statistically significant, and the estimated risk group effect is also almost identical across the methods, with a risk ratio of approximately 1.13 and a p-value close to 0.10. These findings suggest that the presence of tied survival times has little impact on the estimated covariate effects in this dataset.

c) We fitted an accelerated failure time model with a Weibull distribution to the survival data. Let T_i denote the survival time for subject i . The model assumes that T_i follows the Weibull distribution with shape parameter α and scale parameter θ_i ($T_i \sim \text{Weibull}(\alpha, \theta_i)$). The linear predictor is given by: $\eta_i = \beta_0 + \beta_1 \times \text{treatment}_i + \beta_2 \times \text{riskgp}_i$

The scale and shape parameter is defined as: $\theta_i = \exp(-\eta_i)$

The survival time can be written as: $T_i = \exp(\eta_i) \times T_{0i}$ where $T_{0i} \sim \text{Weibull}(\alpha, 1)$ representation shows T_i is obtained by multiplying a baseline survival time, T_{0i} , by a factor depending on the covariates.

Then, the corresponding survival function of the baseline variable T_{0i} and the survival function of T_i are respectively given by: $S_0(t) = \exp(-t^\alpha)$

$$S(t) = \exp\{-[t * \exp(-\eta_i)]^\alpha\} = S_0[t * \exp(-\eta_i)]$$

The R code generated following results:

```
> coef(DRS_AFT_model)
      (Intercept) treatmentplacebo      riskgp
      6.2044391      -0.6446816      -0.1437643

> confint(DRS_AFT_model)
              2.5 %      97.5 %
(Intercept)  4.3848880  8.02399013
treatmentplacebo -1.1651476 -0.12421559
riskgp        -0.3172828  0.02975425
```

The intercept estimate, β_0 , is 6.204 with a 95% confidence interval from 4.385 to 8.024. The estimated coefficient for treatment placebo versus reference group, β_1 , is -0.645 , with a 95% confidence interval from -1.165 to -0.124 . The confidence interval for β_1 does not include zero, indicating a statistically significant treatment effect at the 5% level.

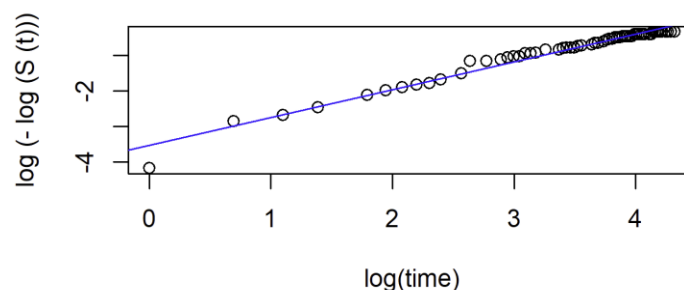
The estimated coefficient for risk group, β_2 , is -0.144 , with a 95% confidence interval from -0.317 to 0.030 , since this interval includes zero, we fail to reject the null hypothesis $H_0: \beta_2 = 0$ at the 5% significance level, indicating insufficient evidence of an association between risk group and survival time.

To assess whether the Weibull accelerated failure time (AFT) model provides a reasonable fit to the data, the adequacy of the assumed Weibull baseline distribution was examined. Taking the logarithm of $S(t)$ twice yields:

$$\log \{-\log [S(t)]\} = \alpha * \log(t) - \alpha * \eta_i$$

This expression implies that, if the Weibull accelerated failure time model is appropriate, a plot of $\log \{-\log [S(t)]\}$ against $\log(t)$ should be approximately linear with slope equal to the Weibull shape parameter α . To assess these properties, Kaplan–Meier survival estimates were plotted on a log–log scale survival function of against log time.

The R code generated the following plot:



The figure displays this plot along with a fitted straight line with the points exhibiting a clear and approximately linear trend across most of the time range. Overall, this diagnostic suggests that the Weibull distribution provides a reasonable fit to the data.